

GENERAL SYSTEMS THEORY

~~Systems Analysis & Design~~

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FRANCISCO JOSÉ DE CALDAS

Outline

- 1 Basic Concepts ✓
- 2 Information Theory and Entropy ✓
- 3 Graphs and Networks Theory ✓
- 4 Paradigms Supporting GST ✓



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What is General Systems Theory?

- General Systems Theory (**GST**) is an interdisciplinary framework for understanding and analyzing complex systems.
- It was introduced by **Ludwig von Bertalanffy** in the 1940s.
- GST focuses on the interconnections and interdependencies between components of a system.
- It is widely applied in fields such as biology, engineering, economics, and social sciences.

biology
mathematics
physics
chemistry
philosophy



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Timeline of General Systems Theory I

- **1928:** Ludwig von Bertalanffy first proposes **organismic theory** as foundation for **GST**.
- **1940:** Bertalanffy publishes "*Der Organismus als physikalisches System*" introducing **open systems** concept.
- **1945:** Bertalanffy formally introduces **General Systems Theory** at University of Chicago.
- **1947:** **Norbert Wiener** coins the term "**Cybernetics**" at **Macy Conferences**.
- **1948:** Wiener publishes "*Cybernetics: Or Control and Communication in the Animal and the Machine*."
- **1948:** Claude Shannon publishes "*A Mathematical Theory of Communication*" founding **information theory**.

Feedback

neural networks



Timeline of General Systems Theory II

Perception

- **1954:** Society for General Systems Research founded by Bertalanffy, Kenneth Boulding, and others.
- **1956:** Jay Forrester develops *Systems Dynamics* at MIT, focusing on feedback loops and time delays.
- **1968:** Bertalanffy publishes *General System Theory: Foundations, Development, Applications*. → mathematical
- **1972:** Donella Meadows et al. publish *The Limits to Growth* using systems dynamics modeling.
- **1975:** Fritjof Capra publishes *The Tao of Physics*, bridging systems thinking and quantum physics.
- **1976:** Richard Dawkins publishes *The Selfish Gene*, applying systems concepts to evolutionary biology.



Timeline of General Systems Theory III

- **1977:** Ilya Prigogine wins Nobel Prize for work on dissipative structures and self-organization.
- **1980s:** Emergence of complexity science and chaos theory, heavily influenced by GST principles.
- **1984:** Santa Fe Institute founded to study complex adaptive systems.
- **1987:** Donella Meadows et al. publish Beyond the Limits, updating systems models.
- **1990:** Peter Senge publishes The Fifth Discipline, popularizing systems thinking in management.
- **1990s-2000s:** GST influences network theory, agent-based modeling, and social network analysis.
- **2004:** Donella Meadows publishes Thinking in Systems: A Primer (posthumously).



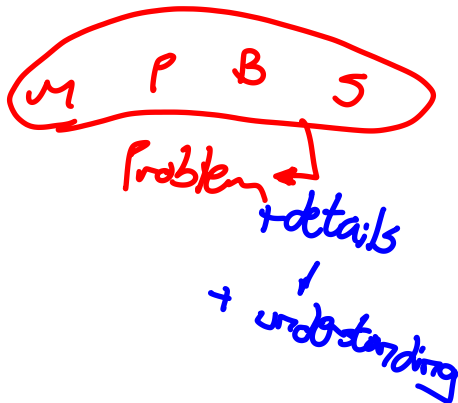
A.N.N.

A.I.

robotics



General Systems Theory I



- In **general systems theory**, the **idea** is to see a **problem** from different autonomous **study areas**; it helps to create a **better full picture** of a problem or situation.
- **Systems** are **dynamic**; for that reason, you need to define **boundaries** and **constraints** to **control analysis**. Also, some systems are **highly susceptible** to changes from the environment.

↓
chaos



General Systems Theory II

- **Ludwig von Bertalanffy**

started to write a book, but he only had some ideas and insufficient background to develop them.

- He waited (twenty years) for mathematical and computational concepts to evolve, and then he was able to finish the book citing more applied concepts.

Alan Turing



General Systems Theory III

- Dynamical Systems*
complex
- In nature in the real world, **everything is a system**. However, the deeper you dive to understand the problem, the more complexity arises.
 - In this context, **systems theory** is useful. Some **patterns** can be detected, and some **details** can be discarded.
abstraction
- classic A.I.*
machine learning

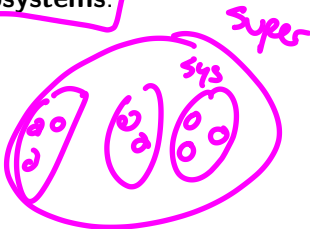


General Systems Theory IV

- **Systems hierarchies** are useful to split big problems into components, work on specific components, and then connect them as the context requires.
- A **system** can be represented by multiple internal systems. The big system is called a **super system**; internal ones are called **subsystems**.

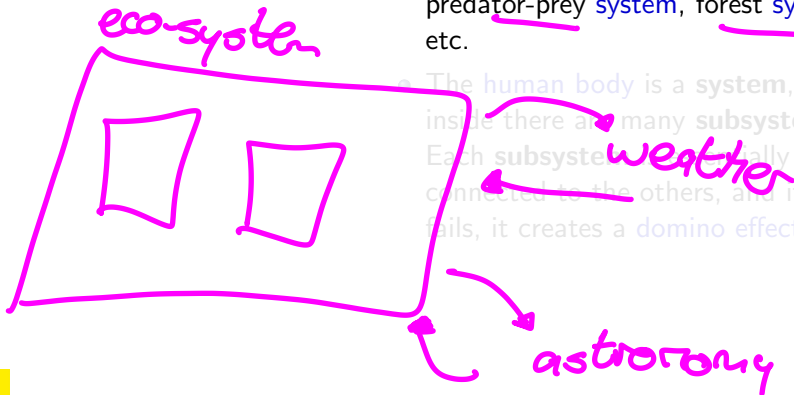
top-down → C.O.P.
architectures

responsibilities segregation



General Systems Theory V

- In nature you could think an ecosystem is a **super system** composed of different **subsystems**:
water system, solar system, predator-prey system, forest system, etc.



- The human body is a system, and inside there are many subsystems. Each subsystem is usually connected to the others, and if one fails, it creates a domino effect.



General Systems Theory V

The Big Short
 → economic crisis 2007~2009
 Dr. House

- In nature, you could think an **ecosystem** is a **super system** composed of different **subsystems**: water **system**, solar **system**, predator-prey **system**, forest **system**, etc.
- The **human body** is a **system**, and inside there are many **subsystems**. Each **subsystem** is essentially connected to the others, and if one fails, it creates a **domino effect**.

medicine



General Systems Theory VI

- **Everything** in the real world, in the universe **is a system**. The hardest thing is to get the right representation.

∇f
 f'
—

- Remember **concepts** such as: snowball effect, butterfly effect, and domino effect. The message is the same: be careful with **details**, failures, and changes; there is **no small impact**.



General Systems Theory VI

- **Everything** in the real world, in the universe, **is a system**. The hardest thing is to get the **right representation**.
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Chaos theory



General Systems Theory VII

- Sometimes you have an ^{Not} expected output. Most of the time, it's hard to achieve, so you must be prepared for everything.

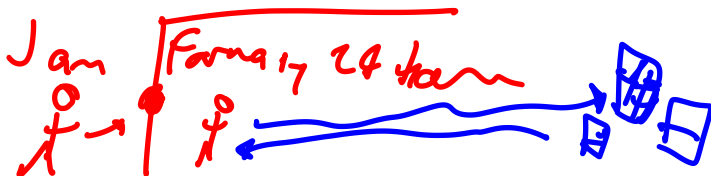
- A black box is a type of model used when you want to get the desired output based on specific input, but you don't want to expose the process used to achieve it.
↓
emergent behaviors



General Systems Theory VII



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- A **black box** is a type of model used when you want to get the **desired output** based on **specific input**, but you don't want to expose the **process** used to achieve it.



General Systems Theory VIII

- **White-box** models are those where the **processes** are open for checking, validating, and following step by step. They are useful when you want to understand **how** the **system works**.

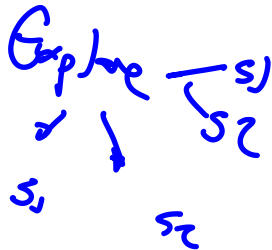


- **Critical Systems Practice** is a methodology for addressing **Critical Systems Thinking**. CSP has 4 main stages: *Explore*, *Produce*, *Intervene*, and *Check* — *EPIC*.



General Systems Theory VIII

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What is Information Theory?

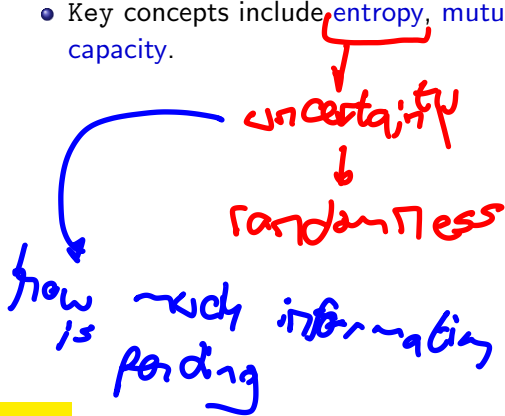
- **Information theory** studies the **quantification**, **storage**, and **transmission** of information.
- It was founded by **Claude Shannon** in 1948.
- Key concepts include **entropy**, **mutual information**, and **channel capacity**.

metrics → mathematics



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Entropy in Information Theory

- **Entropy** measures the uncertainty or randomness in a system.
- **High entropy**: more uncertainty → less predictability.
- **Low entropy**: less uncertainty → more predictability.
- Formula: $H(X) = - \sum p(x) \log_2 p(x)$, where $p(x)$ is the probability of event x .



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Applications of Entropy

- **Data Compression:** Reducing file sizes by removing ~~redundancy~~.
- **Cryptography:** Ensuring secure communication by maximizing entropy.
- **Systems Analysis:** Measuring the complexity and uncertainty of systems.
- **Thermodynamics:** Understanding energy distribution in physical systems.

verano → ↓

los

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What is Graph Theory?

- **Graph theory** studies the **relationships** between **nodes** (*vertices*) and **edges** (*connections*).
- It was introduced by **Leonhard Euler** in **1736**.
- Applications: **social networks**, **transportation systems**, **computer networks**.



Key Concepts in Graph Theory

- **Node (Vertex):** A **point** in the graph.
- **Edge:** A **connection** between two nodes.
- **Degree:** The **number of edges** connected to a node.
- **Path:** A **sequence of edges** connecting nodes.
- **Cycle:** A path that **starts and ends** at the **same node**.



What is Network Theory?

- **Network theory** extends graph theory to study **real-world systems**.
- Focuses on **structure**, **dynamics**, and **functionality**.
- Examples: **Internet**, **power grids**, **biological networks**.



Case Study: Metabolic Network

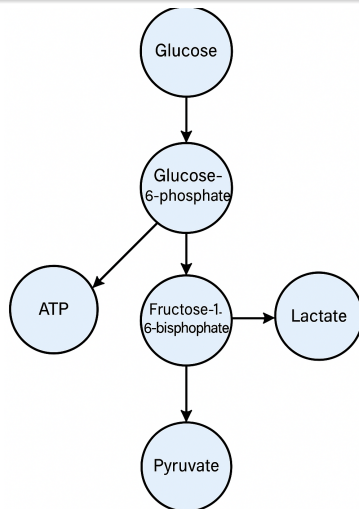


Figure: Metabolic network of *glycolysis pathway*.



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Key Paradigms in General Systems Theory

- **Cybernetics:** Studies **control** and **communication** in systems.
- **Systems Thinking:** Focuses on **interconnections** and **feedback loops**.
- **Complexity Science:** Explores **emergent behavior** in complex systems.
- **Systems Dynamics:** Models **time-dependent** behavior of systems.
- **Agent Theory:** Studies **individual agents** and their interactions in systems.
- **Network Theory:** Analyzes **relationships** and **connections** in systems.
- **Cellular Automata:** Models **discrete systems** with simple rules.
- **Fuzzy Logic:** Deals with **uncertainty** and **imprecision** in systems.
- **Chaos Theory:** Studies **sensitive dependence** on initial conditions.
- **Game Theory:** Analyzes **strategic interactions** between agents.



Cybernetics and GST

- It was founded by **Norbert Wiener** in 1948.
- Focuses on **feedback**, **control**, and **adaptation**.
- *Applications*: **robotics**, **artificial intelligence**, **management systems**.



Systems Thinking and GST

- It emphasizes the **holistic** understanding of systems.
- Key principles: **interdependence**, **feedback**, **emergence**.
- Applications: **organizational management**, **ecology**, **policy-making**.



Complexity Science and GST

- Studies **non-linear**, **adaptive**, and **emergent** systems.
- Examples: **ecosystems**, **financial markets**, **social systems**.
- Tools: **agent-based modeling**, **network analysis**.



Systems Dynamics and GST

- It was developed by **Jay Forrester** in the 1950s.
- Models **feedback loops** and **time delays**.
- Applications: **supply chain management**, **urban planning**, **climate change**.



Agent Theory and GST

- It studies **individual agents** and their **interactions**.
- Key concepts: **autonomy**, **adaptation**, **learning**.
- Applications: **multi-agent systems**, **social networks**, **game theory**.



Network Theory and GST

- It studies **relationships** and **connections** in systems.
- Key concepts: **nodes**, **edges**, **centrality**.
- Applications: **social networks**, **transportation systems**, **biological networks**.



Cellular Automata and GST

- It models **discrete systems** with simple rules.
- Key concepts: **cells, states, neighborhoods**.
- Applications: **pattern formation, biological systems, computer graphics**.



Fuzzy Logic and GST

- It deals with **uncertainty** and **imprecision**.
- Key concepts: **fuzzy sets**, **membership functions**, **fuzzy rules**.
- Applications: **control systems**, **decision-making**, **pattern recognition**.



Chaos Theory and GST

- It studies **sensitive dependence** on initial conditions.
- Key concepts: **chaotic systems**, **bifurcations**, **strange attractors**.
- Applications: **weather prediction**, **financial markets**, **biological systems**.



Game Theory and GST

- Studies **strategic interactions** between agents.
- Key concepts: **players**, **strategies**, **payoffs**.
- Applications: **economics**, **political science**, **biology**.



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

